OLS review.

1. How to find OLS solution $\hat{\beta}$?

$$h(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 = (y - X\beta)^T(y - X\beta)$$
$$= y^Ty - 2\beta^TX^Ty + \beta^TX^TX\beta$$
$$\text{or} \quad = \|y - X\beta\|_2^2$$

To minimize $h(\beta)$

$$\frac{\partial h(\beta)}{\partial \beta} = -2X^Ty + 2X^TX\beta = 0 \qquad \text{(Linear algebra review 4.3-4.4)}$$
$$X^TX\beta = X^Ty$$

X^TX is not invertible when $\text{rank}(X) < p$

2. Theorem 1. $\text{rank}(X) + \text{nullity}(X) = p$

$\text{nullity}(X)$ = dim. of the nullspace of X
$$= \{\nu \mid X\nu = 0\} \qquad \nu \in R^n$$

If $\text{rank}(X) < p$, $\text{nullity}(X) > 0$

i.e, $\text{nullspace}(X) \neq \phi$

let $\eta \in \text{nullspace}(X)$, then $X\eta = 0$ (by definition)

$\Rightarrow \hat{\beta} + \eta$ is also a solution of OLS problem.
if $\hat{\beta}$ is the OLS solution

proof: $(X^TX)(\hat{\beta} + \eta) = X^TX\beta + X^TX\eta$
$$= X^Ty + 0$$
$$= X^Ty$$

For some t,

3. β & $\beta + \eta$ could have opposite sign $\Rightarrow$ interpretation ?
4. prediction is not unique.
5. Fitted values are unique since $X\hat{\beta} = HY$

↓ depend on H (or X) only

where $H = X(X^TX)^*X$ (pseudoinverse)

and $X(\hat{\beta}+\eta) = X\hat{\beta} + X\eta$
$$= X\hat{\beta}$$

Ridge solution (slide 19)

$$h^R(\beta) = \frac{1}{2}\|y - X\beta\|^2 + \lambda\|\beta\|^2$$

$$= \frac{1}{2}(y - X\beta)^T(y - X\beta) + \lambda\beta^T\beta$$

$$\frac{\partial h^R(\beta)}{\partial \beta} = \frac{1}{2}\left(-2X^Ty + 2X^TX\beta\right) + 2\lambda I\beta = 0$$

$$\Rightarrow \quad -X^Ty + X^TX\beta + 2\lambda I\beta = 0$$

$$(X^TX + 2\lambda I)\beta = X^Ty$$

$$\beta^R = (X^TX + 2\lambda I)^{-1}X^Ty \quad (1)$$

prove $(X^TX + 2\lambda I)$ is invertible if $\lambda > 0$.

(X^TX) is a non-negative matrix

i.e. all eigenvalues δ of X^TX are non-negative

let u_δ be eigenvectors, δ be eigenvalues of X^TX

$$(X^TX + 2\lambda I)\cdot u_\delta = (X^TX)u_\delta + 2\lambda I u_\delta$$

$$= \delta u_\delta + 2\lambda I u_\delta$$

$$= (\delta + 2\lambda)u_\delta$$

when $\lambda > 0$ $\quad \delta + 2\lambda > 0$

$\Rightarrow$ $(X^TX + 2\lambda I)$ is invertible

Special case $X^TX = I_p$ (orthogonal case)

ridge solution (1) becomes

$$\beta^R = (I_p + 2\lambda I_p)^{-1}X^Ty$$

$$= (1 + 2\lambda)^{-1}X^Ty$$

OLS solution $\beta^{OLS} = (X^TX)^{-1}X^Ty = X^Ty$

so $\beta^R = \dfrac{\beta^{OLS}}{1 + 2\lambda}$

Lasso solution (Slide 20)

$$h^L(\beta) = \frac{1}{2}(y - X\beta)^T(y - X\beta) + \lambda \sum_j |\beta_j|$$

assume $\beta_j \neq 0$

$$\frac{\partial h^L(\beta)}{\partial \beta_j} = -X^{(j)T}(y - X\beta) + \lambda \, sign(\beta_j) \qquad (1)$$

$X^{(j)}$ is j^{th} column of X

$$sign(\beta_j) = \begin{cases} 1 & \text{if } \beta_j > 0 \\ -1 & \text{if } \beta_j < 0. \end{cases}$$

let $G_j(\beta) = -X^{(j)T}(y - X\beta)$

$$\frac{\partial h^L(\beta)}{\partial \beta_j} = 0 \implies G_j(\beta) = -sign(\beta_j)\lambda$$

Special case $X^TX = I$ (orthogonal)

$$(1) \implies X^{(j)T}y - \underbrace{X^{(j)T}X}_{"I"}\beta = \lambda \, sign(\beta_j)$$

$$\hat{\beta}_j^{OLS} - \beta_j^L = \lambda \, sign(\beta_j)$$

$$\beta_j^L = sign(\beta_j^{OLS})(|\beta_j^{OLS}| - \lambda)_+$$

$\beta_j \neq 0$